

LEAD (Casey, 2026-08-13) — the Millennium work should be reframed around COMPOSITE numbers and GAPS, not PRIMES

Filed by Keeper for the scheduled Millennium proof review. This is Casey's steer, expressed as faithfully as I can; the review takes it as a working hypothesis, not a settled thesis.

The steer, in one line

“I think ‘prime’ is the wrong concept. The work should be about composite numbers and the ‘gaps’ — similar to spectral gaps, where we show the continuum can’t reach without counting.” — Casey

What Casey is correcting

Several current Millennium attempts lean on primality as the load-bearing idea: - $g = 7$ prime (“spectral integrity: the genus must be prime or the lattice factorizes”), Mersenne $2^g - 1 = 127$, the Hamming/perfect-code story, $N_{\max} = 2^g + N_c^2 = 137$ prime, the BSD “all $\binom{g}{k}$ odd via Lucas” route, etc.

Casey’s claim: primality is the wrong organizing concept. It reads as a numerological property of a number, and it’s exactly the kind of thing that invites the “why this number” objection. The organizing concept should instead be:

1. Composite structure — the number’s factorization / how it is *built from parts*, not whether it resists building. A composite carries a counting structure (its factors, its divisor lattice); a prime is defined by the *absence* of one. BST is a theory of *how things are assembled from the five integers* — composites are the native objects; primes are the anomalies.
2. Gaps as spectral gaps — the real phenomenon is that a continuum cannot reach a value without a discrete counting step. This is literally the Yang-Mills mass gap: the scale-free (continuum) flat $\mathbb{R}^4$ theory *cannot* produce the gap; you need curvature, which is counting/discreteness. The gap is the place where the continuum runs out and counting takes over.

The unifying picture (the review’s job to test)

Reframe each Millennium problem as “the continuum can’t cross this gap without counting”: - Yang-Mills — the exemplar, already in hand: no scale-free continuum manifold has a mass gap (the $\mathbb{R}^4$ no-go, Weyl criterion); curvature = the counting that opens the gap. *This is the template.* - Riemann Hypothesis — the zeros cannot leave the critical line: a gap the continuum (off-line region) cannot enter. Recast the temperedness/forcing argument as “the continuum of off-line positions is unreachable without a discrete (counting) obstruction.” - $P \neq NP$ — a complexity gap the polynomial continuum cannot cross: the counting step (witness / parity erasure) is what the continuum of polynomial extensions cannot reach. - Four-Color — the forced-fan / pigeonhole step is a counting obstruction; the continuum of colorings can’t avoid it. - BSD, Hodge, Navier-Stokes — test whether each has a “continuum-can’t-reach-without-counting” spine hiding under the current primality/definitional framing.

Why this is the right BST frame (connections to the corpus)

- Casey’s Principle: entropy = force = counting; Gödel = boundary = definition. The gap is exactly where counting becomes *forced* — the boundary between what the continuum reaches and what only counting reaches.
- The Curvature Principle (“you can’t linearize curvature”): curvature is the discrete/counting content the continuum (linear, scale-free) cannot capture. The mass gap is the first clean instance; the reframe says the others are instances too.
- Limits are lossy (integrals preserve, limits destroy): the *continuum limit* is precisely where the counting information is destroyed — so “the continuum can’t reach it” is the same statement as “the limit threw away the count.”

- Discrete-first (derive discrete, recast continuous onto discrete): the reframe is discrete-first applied to the Millennium problems — stop asking the continuum to reach the answer; show the count it can't reach without.

★ Casey's frame (2026-08-13): the MP *questions* are the problem, not BST — two critiques + the honest output shape

Casey: “*The MPs suffer from sophistication bias and look for general solutions where specific advancements are needed. The wording of the questions is suspect at best... geometric forcing (from D_IV⁵) may provide specific solutions or constraints, yet the overall MP questions should be updated.*” Two distinct critiques, unequal force:

1. Generality bias (all seven). The MPs demand *universality* — all zeros, all curves, all varieties, all problems asymptotically. Prestige flows to the most general statement; but “most general” $\neq$ “most fundamental.” If the remaining problems share one structure (1/rank / gap / composite-vs-continuum), the right advance is specific geometric forcing on the class where the structure lives, not a universal proof. BST is a *forcing* engine, not a *universality* engine.
2. Wording suspect (uneven). Lands HARD on YM (the prize presupposes a rigorous 4D quantum YM that *does not exist* — proving a property of an absent object; the no-go says the presupposition is the trouble), MODERATE on P $\neq$ NP (asymptotic worst-case is a contested modeling choice), SOFT on RH (crisp, well-posed — route RH/BSD/Hodge through critique 1, not 2).

The unifying reframe (= the composite/gaps idea): the MPs are posed as *verification* questions (“*is it true that all...*”); BST *forces* (“*what makes it necessary that...*”). The honest update to each is the turn from verification $\rightarrow$ forcing (RH: “are all zeros on the line” $\rightarrow$ “*what forces* the line”). Same move as composite-over-prime, gap-over-continuum: the MP asks for the continuous universal answer; BST supplies the discrete forcing the continuum-limit framing discarded.

KEEPER LINE (the discipline that keeps this from being sour grapes): “the question should be updated” earns credibility *ONLY* when it *flows from a specific forcing result*, NEVER when it *precedes* one. YM's no-go earns “the R⁴ premise is questionable” because it is a theorem; a bare “the question is wrong, here's our different answer” is goalpost-moving and a referee smells it. Order: specific advance FIRST, question-critique as its consequence.

Output shape for the review — per MP, three lines: (i) the geometric forcing (from D_IV⁵), (ii) the constraint / no-go it proves, (iii) the specific way the question should be updated (verification $\rightarrow$ forcing / general $\rightarrow$ class-specific). This sidesteps both over-claim (“solved”) and under-claim (“failed”), and is stronger than claiming a solution.

What the review should do with this

1. For each of the seven, ask: is the real content a spectral-gap / continuum-can't-reach-without-counting statement? If yes, rewrite the spine that way and see if the “definitions to overcome” (Casey: “the others have definitions to overcome”) dissolve into a clean gap statement.
2. Demote the primality language wherever it is load-bearing to a *consequence* rather than a *cause* — e.g., $g = 7$ being prime is a fact, but the load-bearing thing is the composite/gap structure it sits in, not the primality itself.
3. Keep the honest tiers from K940 / this morning's Guide sweep: YM + 4-color + NS “move the needle”; RH / BSD / P $\neq$ NP / Hodge “have definitions to overcome.” The reframe is aimed squarely at those definitions.

The steer reaches past the Millennium proofs — into the uniqueness argument (Keeper, added 2026-08-13)

While sweeping the Guide, Keeper found the reframe touches the forcing chain that selects D_IV⁵ itself (Vol2_Framework/Ch01_Foundations, Step 3 / T953). Two of the five minimum-requirement conditions rest on primality: - Condition 3: “the Bergman genus must be prime, or the spectral structure factorizes into sub-lattices” (also mislabels $g=7$ as the “Bergman genus” — settled convention: genus = $n_C = 5$; $g = 7$ is the signature).

- Condition 4: “N_max must be prime — a composite N_max decomposes the fine-structure constant into interfering sub-channels.”

Both are secretly gap/indecomposability conditions, with primality as a symptom, not a cause: - Condition 3 → “the spectral lattice must not factorize” (indecomposability). Primality of g is *sufficient* for that but is not the content — the content is that the continuum can’t split the lattice without a counting obstruction. - Condition 4 → “the fine-structure channel must not decompose into interfering sub-channels.” Again a non-factorization / gap statement; N_max prime is the consequence.

So the review should test whether rewriting conditions 3 and 4 as composite/gap (non-factorization) conditions (a) removes the “why must these be prime” objection from the uniqueness argument, and (b) is actually the *truer* statement of what the geometry requires. This would strengthen the T953 uniqueness result the same way the reframe strengthens the Millennium proofs — by naming the gap, not the primality. (Do NOT edit these unilaterally: they intersect Grace’s catalog-owned $g=7$ -genus relabel and are review-level.)

★ The sharpest form (Cal §7541 independently reached this 2026-07-15; Keeper connected it 2026-08-13)

Cal, catching why “ $27 \times 5 + 2 = 137$ ” failed: “*Decomposing a prime — and a prime has no forced decomposition... a prime wants ONE irreducible count — a single spectral cap, a single Verlinde-type divisor, a single irrep dimension — NOT product-plus-remainder.*” This is Casey’s steer reached from the other side, and it gives a concrete, testable prescription:

The rule: - Composites → derive by factorization into the five integers (assembly / eigenvalue products / K-type gradings). BST’s native, abundant, correct mode (sporadic-group orders, Stirling numbers, meson masses all factor into the five integers — “combinatorics $\subseteq$ BST counting”). - Primes → derive as a single indivisible spectral count — a cap / a gap / one irrep dimension / one Plancherel ceiling. NEVER by a composite formula (product + remainder), because a prime has no forced decomposition — any product-plus-remainder that hits it is *one representation among many*, not a derivation. This is exactly “the gap where the continuum can’t reach without counting”: the cap is the single count the continuum can’t exceed.

Immediate consequence for a load-bearing BST claim: the derivation $N_{\max} = N_c^3 \cdot n_c + \text{rank} = 137$ is itself an instance of decomposing a prime. Numerically true, but by Cal’s own principle it is a *consistency check*, not the forced content. The forced content is the spectral cap — T840 derives “no prime > 137 ” from Plancherel-measure decay on D_{IV}^5 , a single spectral ceiling. The review should promote the spectral-cap derivation of 137 (and of g wherever built by formula) to *primary*, and demote the composite formula to a coincidence riding on the cap. Same category test as everywhere: a prime is a gap (one indivisible count), not a sum of parts.

(This also resolves the “three independent routes to 137” tension honestly: if the routes are all product-plus-remainder formulas, they are three coincidental decompositions of one prime, not three derivations — the honest single route is the spectral cap. Verify at the source whether any of the three IS a single-count/cap route; if so, that one is the real derivation.)

Keeper analysis (2026-08-13): WHERE the reframe changes the math vs. the exposition — the review must not conflate these

The steer has two genuinely different consequences depending on the target, and the review should keep them apart or it will over- or under-claim.

Sense A — the uniqueness conditions on g and $N_{\max}$ (T953 cond 3, 4): the reframe is EXPOSITORY, not a new theorem. Here the “gap” is *literal multiplicative factorization of an integer*. A positive integer is prime iff it is multiplicatively indecomposable — so “ $N_{\max}$ prime” and “the α -channel does not decompose into sub-channels” are the same mathematical statement, and likewise “ g prime” $\equiv$ “the spectral lattice does not factor.” The reframe does not weaken or change the condition; it renames it toward its physical content (non-decomposition) and away from the number-theoretic label (primality). *Value*: real but expository — it removes the “why must this number be prime?” objection by revealing that primality was never an extra numerological demand, only the

arithmetic name of “the channel is indecomposable.” Do not sell this as a strengthened theorem; sell it as the honest statement of the existing one.

Source-check RESOLVED (Keeper, 2026-08-13) — cond 3 (g prime) is Sense A, and Lyra’s recast is airtight. The flagged caveat (“is g ’s condition genuinely multiplicative, or Mersenne/code-theoretic?”) resolves to a clean field-theory fact: “ g prime” $\iff$ $\text{GF}(2^g)$ has no proper subfields (a subfield $\text{GF}(2^d) \subset \text{GF}(2^g)$ exists iff $d \mid g$; so the field is subfield-indecomposable exactly when g is prime). So Lyra’s “our error-correcting code can’t be split into smaller codes” = “ $\text{GF}(2^g)$ has no proper subfields” = “ g prime” — an honest rename (Sense A), airtight. *Sharpening for the review*: a second, distinct gap property sits nearby and is NOT what g -primality buys — the Hamming code of length $2^g - 1$ is perfect (its balls tile the space exactly: no gaps, no overlaps), and that perfection comes from the length being $2^g - 1$ (holds for any g), not from g being prime. So cond 3 bundles TWO separable gap statements: field-indecomposability (g prime, Sense A) and perfect packing (length $2^g - 1$, a no-gaps tiling). Both are the reframe in action (indecomposability + exact-packing are both “no hidden substructure”), but they are different properties; the review must name which one the uniqueness argument actually leans on. Cond 4 ($N_{\max}$) is cleanly multiplicative-indecomposable (137 prime $\equiv$ the 137 -mode channel does not split into $a \times b$ sub-channels).

Sense B — the Millennium proofs (RH, $P \neq \text{NP}$, BSD, Hodge): the reframe is SUBSTANTIVE — it recenters the proof. Here the “gap” is a spectral / analytic gap, and primality is genuinely peripheral, often a downstream artifact (e.g. BSD’s “all $\binom{g}{k}$ odd via Lucas” uses $g = 7$ *being* prime, but the load-bearing content is the Chern-hole *counting obstruction* at DOF position 3). Recentring these on “the continuum cannot reach the value without a discrete counting step” is a real change of the proof’s spine, and it is where the “definitions to overcome” (Casey) most plausibly dissolve. This is the substantive win the review should chase — and it must be *earned per problem*, not asserted.

One-line test to classify any given use of a prime in the corpus: *Is the prime doing multiplicative-indecomposability work (Sense A — rename it) or is it a numerological stand-in for a spectral/counting gap (Sense B — rederive the gap and demote the prime)?* Run every load-bearing primality claim through this fork.

Honest status

This is a research lead, not a result. It may sharpen several attempts; it may not fit all seven. The review tests it problem by problem. Tier: LEAD (Casey-originated, unworked). Related: [\[\[feedback_curvature_principle\]\]](#), [\[\[feedback_caseys_principle\]\]](#), [\[\[feedback_discrete_first_pull_continuous_onto_discrete\]\]](#), [\[\[feedback_limits_lossy\]\]](#).

— Keeper, 2026-08-13. Nothing pushed.